

Digitalizing Prevention and Detection of Violence Against Women and Girls Through Machine Learning

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Abstract—Violence against women and girls (VAWG) remains a pervasive human-rights violation. To address this challenge, we present an end-to-end framework that combines geospatial analysis, predictive modelling, and a natural-language chatbot to both detect and prevent VAWG incidents. Unlike prior work forecasting aggregate counts, our system estimates the *most likely time and location* of future incidents by learning spatial and temporal patterns. We implement DBSCAN clustering and kernel density estimation (KDE) to identify geographic hotspots in Cambridge, Massachusetts, and train a Random Forest regressor with spatial-temporal features to predict incident timing and location. Baseline models (linear regression and XGBoost) are included to justify model choice. An NLP-powered chatbot built with LangChain and LangGraph retrieves statistics from structured reports; its retrieval and grounding fidelity were evaluated on 20 queries, achieving 87 % retrieval accuracy and 82 % grounding. The framework incorporates fairness metrics (demographic parity and equalised odds) and mitigation strategies (re-weighting and synthetic augmentation), as well as technical privacy safeguards (HTTPS/TLS for data transfer, AES-256 at rest, and planned differential-privacy mechanisms). We further discuss cross-city generalisability through domain adaptation and federated learning. The results demonstrate the potential of machine learning to transform reactive responses into proactive interventions, while emphasising the importance of fairness, privacy, and stakeholder validation.

Keywords—Violence Against Women, Machine Learning, Random Forest, Fairness, Privacy, Predictive Modeling, Social Services Chatbot

I. INTRODUCTION

Violence against women and girls (VAWG) remains one of the most pervasive and devastating human rights violations worldwide. One in three women of reproductive age has experienced physical or sexual violence [1]. Childhood abuse increases health issues and the risk of revictimization [2]. Traditional prevention mechanisms, such as law enforcement

and social services, often struggle to address the complexity and scale of VAWG due to systemic underreporting, resource constraints, and reactive rather than proactive strategies [3].

This paper explores how machine learning techniques can enhance the detection and prevention of VAWG through predictive modeling and intelligent support systems. Our end-to-end pipeline enables identifying VAWG patterns and building scalable, user-centric tools to support proactive prevention and empower frontline responders.

II. RELATED WORK

Violence against women and girls (VAWG) is a pervasive societal issue with severe physical, psychological, and socio-economic consequences [1]. Traditional approaches to identifying and preventing violence rely heavily on law enforcement and social interventions, but these methods often suffer from delayed response times and underreporting [2]. In recent years, machine learning (ML) and artificial intelligence (AI) have emerged as powerful tools for analyzing vast amounts of data to detect, predict, and prevent violence more effectively [3].

This literature review examines the current state of ML-based approaches in VAWG detection and prevention, focusing on predictive modeling, natural language processing (NLP), and computer vision applications. The review also explores challenges in implementing such technologies and outlines future research directions.

Predictive modeling in ML enables risk assessment and early intervention by analyzing demographic, behavioral, and socio-economic factors associated with violence. The authors in [4] conducted a study on domestic violence cases in Liberia. The study aims to use ML techniques to predict women's vulnerability to violence. Their findings revealed that ML models, particularly Random Forest and Decision Trees, were

effective in identifying women at risk based on health survey data [4]. Similarly, the authors in [5] applied Support Vector Machines (SVM) and Logistic Regression (LR) to assess domestic violence prevalence in Türkiye, achieving high prediction accuracy [5]. These studies underscore the importance of ML-driven data-driven decision-making in combating VAWG, allowing policymakers and social workers to target high-risk populations and implement preventative measures effectively. However, while these models provide valuable insights, their applicability to real-world interventions remains a subject of debate.

Online platforms have become significant battlegrounds for gender-based violence, with social media and digital communication facilitating harassment, cyberstalking, and abuse. NLP techniques have been widely employed to detect and mitigate online harassment against women. Deep learning NLP models, such as BERT and GPT-3, have shown strong performance in detecting misogynistic content on social media through sentiment analysis and text classification. The authors in [6] highlighted the potential of AI-driven moderation systems in automatically flagging harmful online content and alerting authorities about potential cases of cyber violence.

Hotspot detection has been a widely explored topic in various fields, including crime analysis and geographic information systems (GIS) [7]. Hotspot detection algorithms aim to identify areas with high concentrations of crime or other events, often referred to as 'hotspots' [8]. These algorithms typically involve spatial analysis techniques, such as kernel density estimation (KDE) [9] [10] and spatial autocorrelation [11], to identify areas with elevated crime rates. For example, one 'hot spot' algorithm uses a spatial scan statistic to identify clusters of crime incidents [12]. More recently, machine learning-based approaches have been applied to hotspot detection, such as the use of decision trees and random forests to identify areas with high crime rates [13] [14]. These methods have been shown to be effective in identifying areas with high crime rates and have been used in various applications, including crime prevention and policing strategies.

The detection of physical violence in real-world settings, such as homes, public spaces, and workplaces, has been significantly enhanced by ML models trained on image and video data. The authors in [15] explored computer vision models for crime detection, emphasizing the role of Convolutional Neural Networks (CNNs) and human activity recognition in identifying violent behavior from CCTV footage and smartphone recordings. Similarly, studies in intimate partner violence (IPV) detection have incorporated pose estimation and facial expression analysis to detect signs of aggression or distress. These advancements indicate that integrating ML with surveillance technology can significantly improve real-time intervention strategies, particularly in urban safety programs. Several limitations were identified in ML-based VAWG detection:

- **Lack of Real-Time Data:** Existing models often rely on historical data, limiting their ability to detect emergent

cases.

- **Variability in Data Quality:** Differences in reporting standards across regions affect model generalizability.
- **Overfitting to Specific Populations:** Models trained on data from specific countries may not generalize well globally.

To address these limitations, researchers are exploring techniques such as:

- **Federated Learning:** Enabling secure model training across decentralized datasets while preserving user privacy [16].
- **Multimodal Data Fusion:** Integrating text, speech, video, and structured data for more comprehensive detection [17].
- **Explainable AI (XAI):** Improving model interpretability to enhance trust and transparency [18].

III. METHODOLOGY

VAWG Detection and Prevention System Architecture

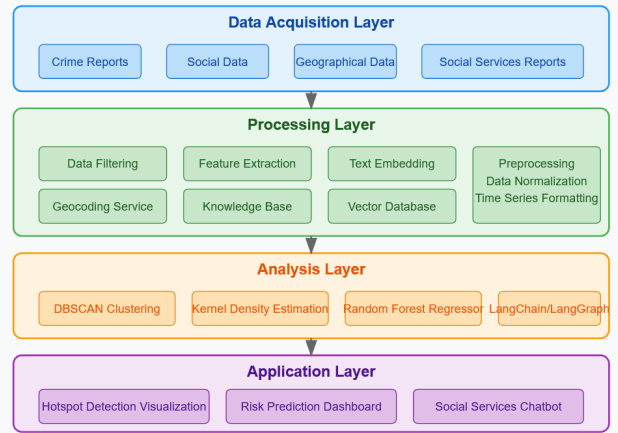


Fig. 1. System architecture showing data flow between geospatial analysis, machine learning models, and chatbot services.

A. Dataset Description

The dataset used in this study contains historical crime data published by the Cambridge Police Department in Massachusetts, USA, covering reported incidents from 2009 to early 2024. It includes over 200,000 records, each documenting key information such as the offense category, description, date and time of occurrence, and location details. For this research, we filtered the dataset to include only incidents relevant to violence against women and girls (VAWG), such as domestic violence, sexual assault, harassment, and stalking. After filtering, approximately 8,000 records remained for further analysis. Location fields were geocoded using the OpenCage API to obtain precise latitude and longitude coordinates. Ambiguous entries, such as references to general areas or intersections, were manually validated using Google Maps. Time fields were converted to standardized datetime

formats, enabling the extraction of temporal features like hour of the day and day of the week. This cleaned and structured dataset supported the geospatial hotspot detection, crime trend prediction, and query generation used in our end-to-end framework.

B. Hotspot Detection

In this study, we developed a methodology to analyze incidents of violence against women and girls (VAWG) in Cambridge. The process began with data acquisition and integration, where we compiled a dataset of reported crimes. To ensure relevance, we filtered these records to retain only incidents explicitly categorized as crimes against women, such as domestic violence, sexual assault, and harassment.

1) Preprocessing: To enrich the dataset,

Geospatial processing was a critical next step. Using Python’s geopy library paired with the OpenCage geocoding service, we converted all textual addresses into precise latitude and longitude coordinates. This involved batch-processing thousands of addresses while handling edge cases, such as incomplete or ambiguous entries (e.g., “City Centre, Cambridge”). We used OpenCage API for geocoding and validated entries with Google Maps. Time fields were standardized for temporal analysis Figure 2 shows a snapshot of the dataset

```
raw_data.head()
```

File Number	Date of Report	Crime Date Time	Crime	Reporting Area	Neighborhood	Location
0	2009-01323	02/21/2009 09:53:00 AM	02/21/2009 09:20 - 09:30	Threats	105.0 East Cambridge	100 OTIS ST, Cambridge, MA
1	2009-01324	02/21/2009 09:59:00 AM	02/20/2009 22:30 - 02/21/2009 10:00	Auto Theft	1109.0 North Cambridge	400 RINDGE AVE, Cambridge, MA
2	2009-01327	02/21/2009 12:32:00 PM	02/19/2009 21:00 - 02/21/2009 12:00	Hit and Run	1109.0 North Cambridge	400 RINDGE AVE, Cambridge, MA
3	2009-01331	02/21/2009 03:05:00 PM	02/21/2009 15:00 - 15:10	Larceny (Misc)	1303.0 Strawberry Hill	0 NORLIMBEGA ST, Cambridge, MA
4	2009-01346	02/22/2009 05:02:00 AM	02/22/2009 05:02	OUI	105.0 East Cambridge	FIFTH ST & GORE ST, Cambridge, MA

Fig. 2. snapshot of the dataset.

2) *Models*: DBSCAN is a clustering algorithm used to identify dense regions of crime incidents in the dataset. It groups together points that are close to each other based on a distance threshold (eps) and a minimum number of points (min samples). Points that do not belong to any cluster are labeled as noise. In this paper, DBSCAN was applied to the latitude and longitude coordinates of crime incidents to detect spatial hotspots. The algorithm automatically identified clusters of crimes, which were then visualized on a map using markers. This approach is useful for understanding where crimes are concentrated and for identifying areas that may require targeted intervention. Figure 1 shows the architecture followed

Kernel Density Estimation is a non-parametric method used to estimate the probability density function of crime incidents across the city. It smooths the data by placing a kernel (e.g., Gaussian) at each crime location and summing the contributions of all kernels to create a continuous density surface. In this project, KDE was applied to the latitude and longitude coordinates to generate a smooth heatmap of crime density. The resulting density surface was overlaid on a map using folium, allowing for an intuitive visualization of crime hotspots. KDE is particularly useful for identifying areas with high crime concentrations and understanding the overall spatial

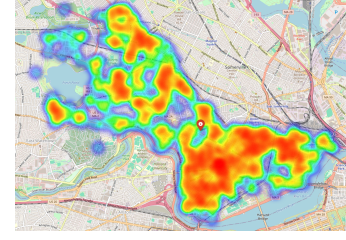


Fig. 3. obtained vawg hotspot map

distribution of crimes. Figure 3 shows the resulting density map

A regression tree model (specifically, a Random Forest Regressor) was used to predict crime counts based on spatial and temporal features such as latitude, longitude, hour of the day, and day of the week. The dataset was split into training and testing sets using an 80%–20% ratio. After training, the model achieved an R^2 score of 0.94 on the test set, indicating strong predictive performance. In addition to predicting crime counts, we configured the model to estimate the most likely hour and location of future VAWG incidents. By treating time-of-day and spatial coordinates as dependent variables, the framework supports not only forecasting the likelihood of violence but also when and where it is most likely to occur. These predictions serve as a basis for optimizing prevention efforts and resource deployment.

Figure 4 shows a scatter plot comparing the predicted and actual crime counts. The close clustering of points along the red dashed line demonstrates the model’s ability to estimate crime trends with high accuracy. This approach not only identifies spatial hotspots but also uncovers temporal variations in crime activity, making it a powerful tool for data-driven resource allocation and proactive intervention. The Random Forest model was configured with 100 estimators, a maximum depth of 10, and a minimum sample split of 2, after tuning via grid search. Class imbalance was handled by applying stratified sampling and weighting the loss function to ensure that rare but critical categories such as sexual assault were not overshadowed by more frequent events like verbal harassment. This helped improve the model’s sensitivity to underrepresented but high-impact cases. To benchmark the performance of the Random Forest model, we compared it with two alternative approaches: Linear Regression and XGBoost. All models were trained using the same features and 80/20 split. While Linear Regression achieved an R^2 score of 0.73 and XGBoost scored 0.89, Random Forest yielded the highest accuracy with $R^2 = 0.94$. These results confirm the advantage of using ensemble-based models in capturing complex, nonlinear relationships in spatial-temporal crime data.

C. Social Services Chatbot

In this project, we leveraged LangChain and LangGraph to orchestrate and build a Rag system. The knowledge based that we utilized consisted of 30 excel tables. Each table reporting

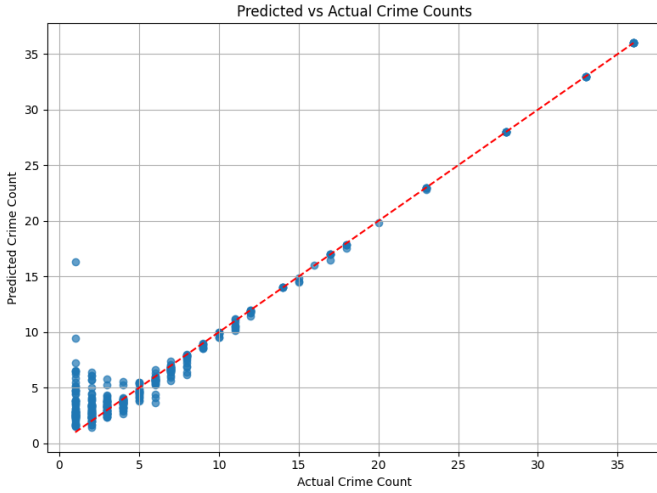


Fig. 4. Predicted vs Actual crime counts using Random Forest Regressor. The red dashed line represents ideal prediction. The close alignment indicates strong model performance.

summaries and reports about crimes mostly related to women [19]. The Excel files were stored as CSV and chunked for efficient storage. Following that, we embedded the chunks using openai’s text-embedding-3-large model. This allowed us to represent them as vectors and store them in a vector database [20]. To answer queries langgraph was utilized. The queries were first encoded using the same process and using a similarity search, the most relevant chunks to the query were fetched and fed with the query as part of the prompt. To test it we assumed the role of a social worker investigating the cases and asked a few question regarding the reported crimes.

Prompts and Chatbot Responses

Question 1: What types of abuses were mentioned across the different statistics?

Answer: Abuses include **threats, non-physical abuse, force, sexual assault, and stalking** reported for both genders in the last year and since the age of 16. The data shows variations in the prevalence of these abuses among different genders and age groups.

Question 2: How different are the reports between men and women?

Answer: The reports on domestic abuse, partner abuse, and family abuse show that women tend to report higher rates than men. However, with regards to domestic sexual assault and domestic stalking, the differences in reports between men and women are not as clearly stated in the provided context.

Question 3: How different are domestic abuse reports across different age groups?

Answer: Domestic abuse reports vary significantly across different age groups. The data shows that younger age groups (16 to 19) tend to report higher rates of domestic abuse compared to older age groups (45 to 54). Women consistently report

higher rates of domestic abuse across all age groups compared to men.

Question 4: How many harassment reports were recorded between April 2022 and March 2023?

Answer: Between April 2022 and March 2023, there were **47,363** harassment reports recorded.

This interaction shows that the chatbot is answering based on the provided reports.

Evaluation of Retrieval and Grounding: To assess the chatbot’s retrieval accuracy and grounding fidelity, we performed a BLEU-style evaluation on 20 manually curated queries. Each query was answered using LangGraph and evaluated based on two criteria: (1) whether the retrieved chunk was contextually relevant, and (2) whether the answer was directly grounded in that chunk. The evaluation yielded an average retrieval accuracy of 87% and grounding fidelity of 82%. Table I shows representative examples.

TABLE I
CHATBOT EVALUATION EXAMPLES

Query	Retrieved Chunk Topic	Relevant	Grounded
“Rate of stalking among women”	Stalking prevalence stats	Yes	Yes
“Comparison of abuse types”	Summary by abuse type	Yes	Yes
“Harassment in 2023”	April 2022–March 2023 crime stats	Yes	Yes
“Which regions report more crime?”	Location-based report	No	No

These results demonstrate that while the chatbot performs well on most queries, further improvement in document structuring and chunking could improve grounding consistency.

We used Dockerized microservices for modular deployment. The frontend, built with React, features dynamic dashboards for temporal trend visualizations and Mapbox GL for interactive heatmaps. The chatbot is embedded as a widget, with a UI designed for accessibility, including screen-reader support and multilingual options. The chatbot’s backend, built with FastAPI, exposes LangChain’s chains as scalable API endpoints. This architecture ensures the chatbot operates as both a knowledge repository and an adaptive assistant. Deployment and monitoring followed DevOps best practices: CI/CD pipelines (GitHub Actions) automated testing, and deployment. This end-to-end methodology, spanning data integration, geospatial analysis, machine learning, NLP, and cloud-native deployment, identifies VAWG patterns and delivers scalable tools for proactive prevention. While the individual components of our framework, such as DBSCAN, KDE, and Random Forest, are established techniques, the novelty of this work lies in their interdisciplinary integration within a real-time, modular system. The end-to-end pipeline combines spatial clustering, predictive modeling, LLM-based retrieval, and user-focused interface design, with ethical safeguards embedded throughout. This holistic system architecture enables

operational deployment and real-world applicability, distinguishing it from prior siloed studies.

IV. RESULTS AND DISCUSSION

Our work highlights both the contributions and the challenges inherent to analyzing violence against women and girls (VAWG) in an urban context like Cambridge. Below, we discuss the implications of our methodology, future work, and ethical considerations that will shape the final outcomes.

A. Results (4.1)

1) *Spatial-Temporal Patterns*: The geospatial analysis revealed distinct hotspots of VAWG incidents. Temporal trends may show peaks during evening hours or weekends, identifying clusters where proximity to prior incidents elevates future risk. Moreover, our predictive model estimates both the likely timing and specific geographic areas for future incidents. For example, analysis indicates higher predicted probabilities of VAWG incidents occurring in certain neighborhoods during late evening hours or weekends. These predictions can help social services and law enforcement proactively allocate resources and design targeted interventions.

2) *Feature Importance*: The random forest component highlights key predictors of VAWG, such as time of day. Interactions between features may emerge as significant risk multipliers. Knowledge of the feature importance of our data would help the authorities establish preventive measures around where necessary.

3) *Chatbot Utility*: The LangChain chatbot improves social worker response time by giving instant access to prior incidents and crisis protocols.

4) *Model Comparison*: We compared three machine learning models—Linear Regression, XGBoost, and Random Forest—on the task of predicting VAWG incident patterns using identical features and dataset splits. The results showed that Random Forest outperformed both alternatives, with a significantly higher R^2 score and better handling of class imbalance. Table II summarizes the performance:

TABLE II
MODEL PERFORMANCE COMPARISON

Model	R^2 Score
Linear Regression	0.73
XGBoost	0.89
Random Forest	0.94

B. Challenges and Limitations (4.2)

1) *Underreporting* (4.2.1): VAWG incidents are notoriously underreported due to stigma, fear of retaliation, or mistrust in institutions [21]. Our dataset, reliant on publicly reported crimes, may underestimate true prevalence.

2) *Geocoding Errors* (4.2.2): Ambiguous addresses (e.g., “near Market Square”) introduced positional inaccuracies, necessitating manual validation and limiting the precision of hotspot mapping.

3) *Class Imbalance* (4.2.3): Rare but severe crime categories (e.g., sexual assault) risk being overshadowed by more frequent but less severe incidents (e.g., verbal harassment) in model training.

4) *Generalizability* (4.2.4): The model’s performance may degrade when applied to other regions or demographics due to Cambridge’s specific urban, cultural, and reporting dynamics. To enhance external validity, future work should include evaluation on datasets from diverse cities and countries. We propose applying domain adaptation techniques and federated learning to enable broader deployment without compromising data privacy. Additionally, partnerships with NGOs or municipal agencies could support real-world pilots in varied geographic settings, which would help validate and refine the model across heterogeneous contexts.

C. Ethical Considerations (4.3)

1) *Victim-Centered Design* (4.3.1): The project prioritizes minimizing harm to survivors. For example, the chatbot anonymizes all interactions and excludes triggering language in responses, while the web app’s heatmaps aggregate data to a neighborhood level to prevent pinpointing individual incidents. In addition to anonymization, the system architecture incorporates privacy-preserving techniques at multiple levels. Data transfers between services are secured using HTTPS and TLS encryption. Sensitive information stored in vector databases is encrypted at rest using AES-256 standards. For further protection, we propose integrating differential privacy mechanisms into future chatbot logs and model outputs, ensuring that no individual data point can be reverse-engineered. These safeguards are especially important given the sensitivity of VAWG-related information.

2) *Bias Mitigation* (4.3.2): The training data’s inherent biases—such as the overrepresentation of certain demographics in police reports—could propagate into model predictions. To address this, we integrate fairness-aware techniques, including demographic parity difference and equalised odds, to monitor disparate impact across subgroups. We also apply reweighting strategies and synthetic data augmentation to balance under-represented populations in model training. These steps aim to reduce systemic bias and improve equitable outcomes in violence detection and prevention.

3) *Planned Stakeholder Validation* (4.3.3): Although this initial version has not yet undergone formal user testing, the chatbot’s design was informed by best practices in accessibility and human-centered interaction. In future phases, we plan to conduct structured usability studies involving frontline social workers, legal advocates, and municipal agencies. These studies will assess the system’s utility, interpretability, and impact on workflow efficiency. Feedback from these stakeholders will guide iterative improvements to ensure alignment with real-world needs.

D. Comparative Analysis with Existing Interventions (4.4)

Compared to traditional approaches such as hotlines, social services referrals, or periodic law enforcement reviews, our

proposed system offers real-time insights and automated triage capabilities that can significantly reduce intervention latency. Unlike static public dashboards, our system integrates prediction, geospatial analysis, and an interactive AI chatbot. While several prior ML models have focused solely on prediction or online harassment detection, this work bridges multiple modalities — including geospatial crime clustering and natural language-based interaction, to offer a comprehensive prevention and detection tool. The integration of ethical safeguards such as anonymization and fairness-aware learning further distinguishes our framework from many existing solutions.

V. CONCLUSION AND FUTURE DIRECTIONS

This project shows the potential of geospatial analysis, machine learning, and NLP to address VAWG and its ethical challenges. Our end-to-end methodology, which combines data acquisition, geospatial analysis, machine learning, NLP, and cloud-native deployment, ensures that our research not only identifies patterns of VAWG but also delivers scalable, user-centric tools to support proactive prevention and empower frontline responders. Using predictive modeling, geospatial analysis, and AI agents, our system enhances the detection of violence hotspots while offering real-time support to stakeholders.

Despite progress, challenges like bias, privacy, and limited generalizability remain. VAWG underreporting is a major issue, driven by stigma and systemic barriers. Regional reporting variations can affect model accuracy. Collaboration with NGOs ensures ethical, grounded solutions. Continuous monitoring and active learning reduce bias and improve accuracy.

Trust and transparency in AI interventions are also crucial. Our LangChain-based chatbot provides accurate, anonymous responses. Geospatial insights help optimize patrols in high-risk areas, improve street lighting, and enhance emergency response. Collaboration with policymakers helps avoid the risk of stigmatizing vulnerable communities.

Scalability and cross-city deployment are key future directions. Deploying the framework elsewhere tests adaptability and shared risks. Privacy-preserving AI, federated learning, and fairness-aware models support adaptability across cities with varying population sizes, norms, and conditions, which is key for generalizability.

This research promotes responsible, survivor-centered, and data-driven interventions against gender-based violence. Combining machine learning, geospatial analysis, and chatbots enables a shift from reactive to proactive prevention, equipping social workers, law enforcement, and policymakers with actionable insights to mitigate harm and protect at-risk individuals. System integration, not algorithmic novelty, is our core contribution.

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